ASCII MASTER FLOW — PANEL 1/12: Greenhouse-effect firewall

NATURAL EFFECT -> absorbs and re-emits outgoing longwave radiation
FUNCTION -> maintains a habitable energy balance
HUMAN ENHANCEMENT -> raises concentrations and net forcing
CLIMATE RESPONSE -> warming plus wider system changes
EXAM RULE -> preserve the natural-versus-enhanced distinction
MUST REMEMBER: Climate change follows radiative forcing and Earth-system feedbacks from...

ASCII MASTER FLOW — PANEL 2/12: Energy-balance rail

```
SUN -> incoming predominantly shortwave energy
SURFACE -> absorbs part and warms
EARTH -> emits outgoing longwave infrared energy
GREENHOUSE GASES -> absorb and re-emit part of longwave flow
IMBALANCE -> forcing changes energy retained by the climate system
```

ASCII MASTER FLOW — PANEL 3/12: Emission-to-concentration ledger

```
ACTIVITY -> emissions flow during a stated period
ATMOSPHERE -> concentration after sources and sinks interact
ACCUMULATION -> long-lived gases build an atmospheric stock
FLOW CUT -> slows addition but does not erase the stock
REMOVAL -> changes the stock only within a stated boundary
```

ASCII MASTER FLOW — PANEL 4/12: Forcing-feedback fork

```
EXTERNAL OR IMPOSED ENERGY CHANGE -> radiative forcing
CLIMATE RESPONSE TRIGGERED -> feedback
AMPLIFIES INITIAL CHANGE -> positive feedback
DAMPENS INITIAL CHANGE -> negative feedback
SIGN -> direction of response, never moral value
```

ASCII MASTER FLOW — PANEL 5/12: Water and particle matrix

WATER VAPOUR -> greenhouse gas commonly acting as feedback
SULPHATE-LIKE AEROSOLS -> often cooling influence
BLACK CARBON -> absorbing warming influence
SNOW OR ICE DEPOSITION -> can lower albedo
BOUNDARY -> agent, location and mechanism determine the sign

ASCII MASTER FLOW — PANEL 6/12: Gas comparison gate

GWP -> relative integrated influence over a named horizon
POTENCY -> effect per unit mass under the metric
LIFETIME -> persistence in the atmosphere
EMISSION SCALE -> quantity released
POLICY -> near-term and cumulative levers are not identical

ASCII MASTER FLOW — PANEL 7/12: Weather-to-climate ladder

EVENT -> individual weather occurrence
SEASON OR YEAR -> short-period variability
DISTRIBUTION -> frequency, intensity and pattern
LONGER RECORD -> climate trend assessment
RULE -> fluctuation around a trend is not trend reversal

ASCII MASTER FLOW — PANEL 8/12: Attribution evidence chain

```
OBSERVATION -> define the measured change
DETECTION -> distinguish signal from expected variability
DRIVERS -> compare human and natural influences
MODELS PLUS PHYSICS PLUS RECORDS -> converging evidence
ATTRIBUTION -> qualified contribution, not slogan causation
```

ASCII MASTER FLOW — PANEL 9/12: Event-claim firewall

```
DEFINE EVENT CLASS -> place, duration and metric
COMPARE WORLDS -> with and without human influence
ESTIMATE CHANGE -> probability or intensity
RETAIN OTHER DRIVERS -> exposure, vulnerability and variability
NO SOLE CAUSE -> disaster impact is multi-causal
```

ASCII MASTER FLOW — PANEL 10/12: Observation-projection matrix

```
OBSERVATION -> measured past or present
ATTRIBUTION -> explanation of observed change
SCENARIO -> conditional assumptions
PROJECTION -> modelled response under a scenario
FORECAST -> different claim with a prediction horizon
CLOSE DISTINCTION: Weather is not climate, emission flow is not concentration stock,...
```

ASCII MASTER FLOW — PANEL 11/12: Climate-response fork

```
SOURCE REDUCTION -> mitigation
SINK ENHANCEMENT -> mitigation with accounting boundary
EXPOSURE OR VULNERABILITY REDUCTION -> adaptation
RESIDUAL HARM -> separate loss-and-damage question
PORTFOLIO -> mitigation and adaptation are complementary
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State unit, baseline, period, scenario...
```

ASCII MASTER FLOW — PANEL 12/12: Science answer spine

```
DEFINE -> natural effect, human enhancement and energy flow
ACCOUNT -> emissions, concentrations, stock and net balance
EXPLAIN -> forcing, feedback and gas-specific properties
EVALUATE -> trend, detection, attribution and projection
QUALIFY -> scale, scenario, confidence and no invented figures
```